

Riley Wall

1 (204) 230-4017 | rileypennwall@hotmail.com | [linkedin.com/in/riley-wall](https://www.linkedin.com/in/riley-wall) | github.com/BuckForth

Education

UNIVERSITY OF MANITOBA

Bachelor of Science in Computer Science

Winnipeg, MB

Sept 2015 - June 2020

UNIVERSITY OF MANITOBA

Master of Science in Computer Science

Winnipeg, MB

Jan 2022 - June 2025

Experience

SESSIONAL INSTRUCTOR (COMPUTER SCIENCE)

UNIVERSITY OF MANITOBA

Winnipeg, MB

Jan 2022 – Dec 2024

- Prepared lecture materials and lesson plans for university courses covering data structures and AI/ML algorithms.
- Managed teams of assignment markers/graders to ensure graded assignments returned in a timely manner.

POS SUPPORT & TECHNICIAN

COMARK

Winnipeg, MB

July 2019 - Jan 2022

- Assisted in daily remote troubleshooting of “Point of Sale” systems for retail clothing stores across Canada.
- Maintained accurate bug reporting and mitigation strategies to ensure ongoing functionality of various software and hardware components.
- Responsible for sourcing replacement components and delivery to various in store locations.

Projects

Humanoid Robotics, Python, I2C, 3D printing, Raspberry PI, OpenCV

Jan 2020 – Present

- Designed and printed components for assembly.
- Current iteration has 30 degrees of freedom and stands about 18 inches tall.
- Used OpenCV for visual information processing.
- Created an I2C interface using Python to allow smooth servo control.

Unity Development, C#, Unity3D, GPU processing, Computer Modeling

Jan 2020 – Present

- Develop procedurally generated environments.
- Implemented a massively parallel convolute kernel using compute shaders. (Used for CNN learning)
- Simulate large agent-based models in a 3d environment where line of sight and simulated viewpoints can be extracted to generate training data for mobile AI agents.
- Built testing suite to train virtual drones in the task of navigating 3D spaces.

Technical Skills

Languages: Java, Python, C/C++, C#, HTML/CSS

Frameworks: Cuda, Junit, React

Developer Tools: Git, VS Code, Visual Studio, Eclipse, Unity3D

Libraries: TensorFlow, OpenCV, NumPy, Matplotlib